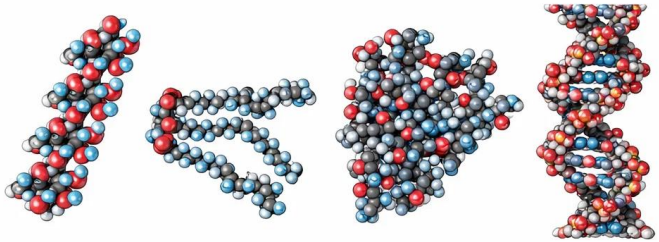
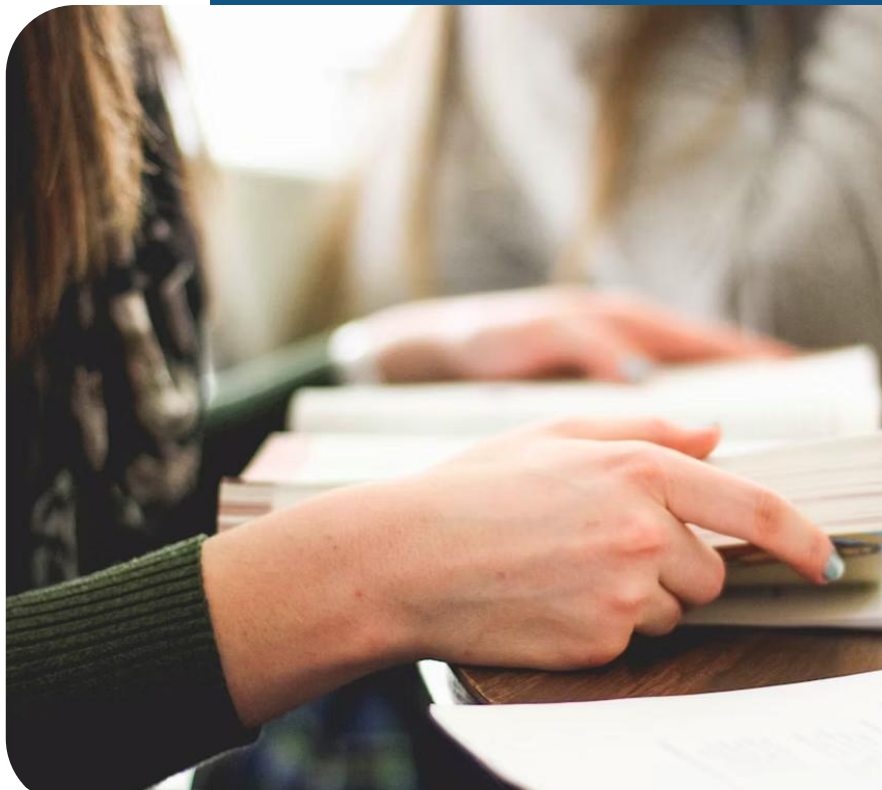
# Biological Macromolecules

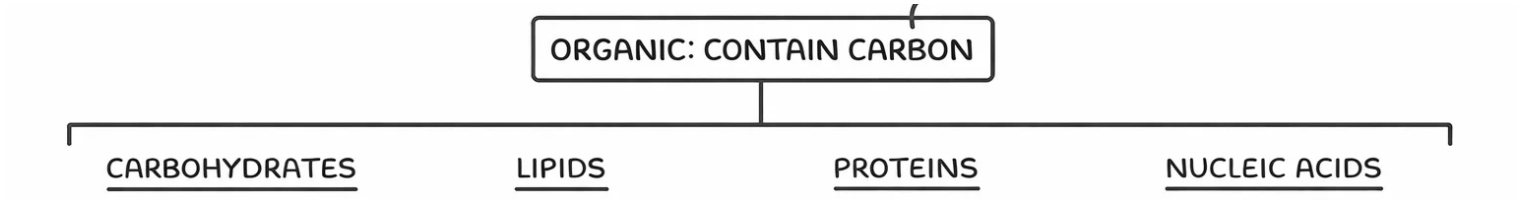


Carbohydrate (starch) | Lipid (triacylglycerol) | Protein (enzyme) | Nucleic acid (DNA)

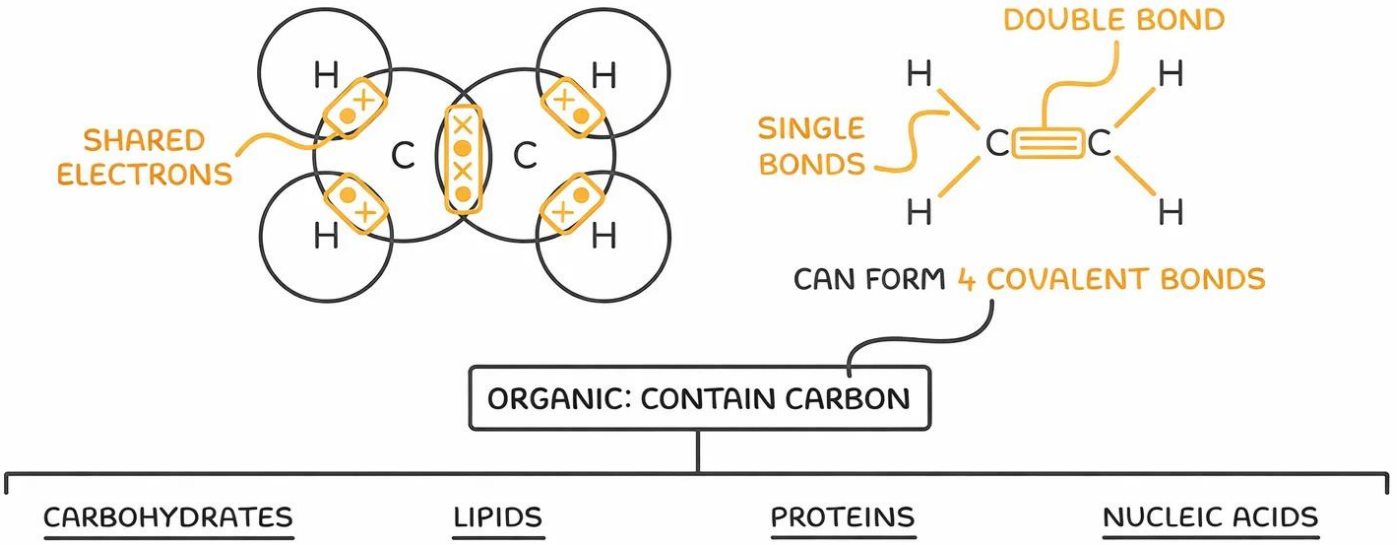
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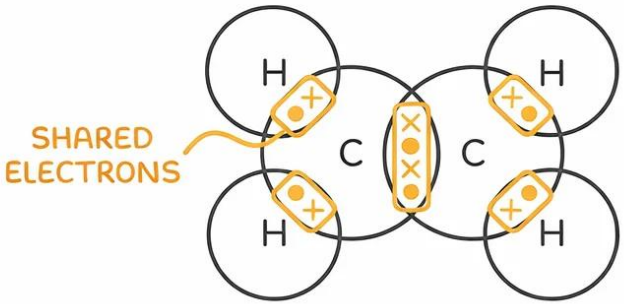
# Organic Molecules

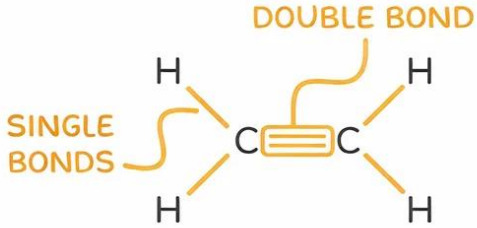


# Organic Molecules



# Organic Molecules



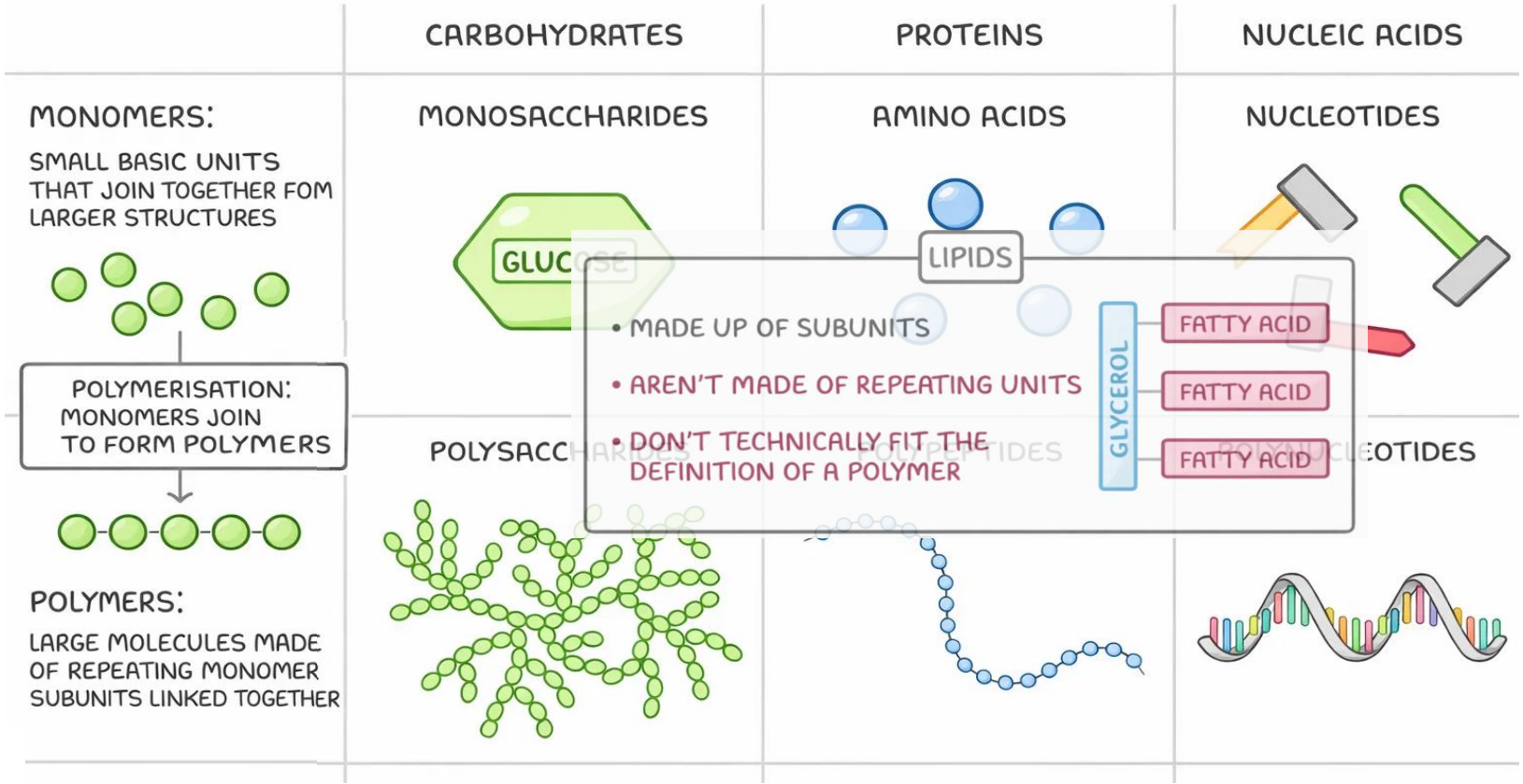




CAN FORM 4 COVALENT BONDS

**ORGANIC: CONTAIN CARBON**

| CARBOHYDRATES | LIPIDS | PROTEINS | NUCLEIC ACIDS |
|---|---|---|---|
| ONLY CONTAIN: | CONTAIN: | CONTAIN: | CONTAIN: |
| • CARBON (C) | • CARBON (C) | • CARBON (C) | • CARBON (C) |
| • HYDROGEN (H) | • HYDROGEN (H) | • HYDROGEN (H) | • HYDROGEN (H) |
| • OXYGEN (O) | • OXYGEN (O) | • OXYGEN (O) | • OXYGEN (O) |
| | | • NITROGEN (N) | • NITROGEN (N) |
| | | | • PHOSPHORUS (P) |

# What Monomers and Polymers are?



# Condensation and Hydrolysis

Also read other key roles of water in life

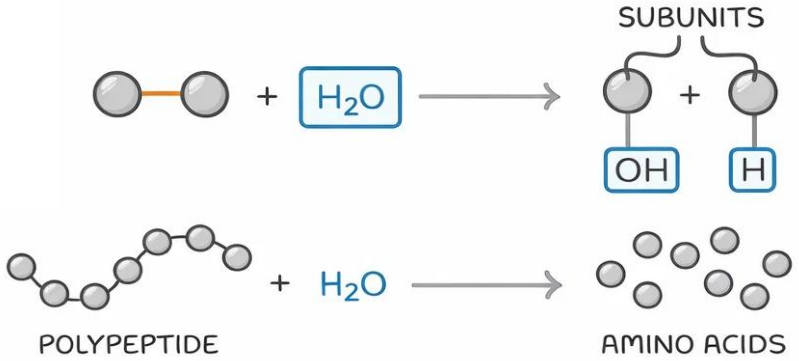
## CONDENSATION REACTIONS:

- HOW MOST POLYMERS ARE SYNTHESISED
- WATER REMOVED TO FORM CHEMICAL BOND
- REQUIRES ENERGY (SUPPLIED BY ATP)




## HYDROLYSIS REACTIONS:

- POLYMERS, OR OTHER LARGE MOLECULES, BROKEN DOWN INTO SUBUNITS
- WATER IS USED TO BREAK A CHEMICAL BOND
- RELEASE ENERGY



# What Carbohydrates Are?

BIOLOGICAL MOLECULES THAT CONTAIN ONLY 3 ELEMENTS: CARBON, HYDROGEN & OXYGEN

GENERAL FORMULA: $C_X(H_2O)_Y$

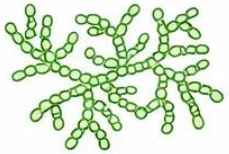
PRESENT IN A 2:1 RATIO (LIKE IN WATER, $H_2O$)

## FUNCTIONS IN LIVING ORGANISMS:

1. SIMPLE CARBOHYDRATES SUPPLY ENERGY DIRECTLY TO CELLS



2. ENERGY STORED AS COMPLEX CARBOHYDRATES (STARCH IN PLANTS & GLYCOGEN IN ANIMALS)



3. FORM STRUCTURAL COMPONENTS (CELLULOSE IN PLANT CELL WALLS & CHITIN IN INSECT EXOSKELETONS)

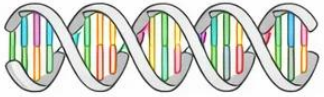




4. INVOLVED IN CELLULAR RECOGNITION (HELP CELLS IDENTIFY & COMMUNICATE WITH EACH OTHER)




5. BUILDING BLOCKS FOR OTHER MOLECULES (DEOXYRIBOSE & RIBOSE NEEDED TO MAKE NUCLEIC ACIDS LIKE DNA & RNA)

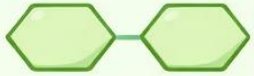


### 3 MAIN TYPES OF CARBOHYDRATE:

1. MONOSACCHARIDES

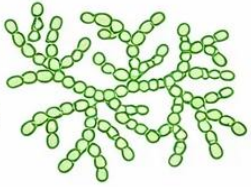


2. DISACCHARIDES



SIMPLE CARBOHYDRATES

3. POLYSACCHARIDES

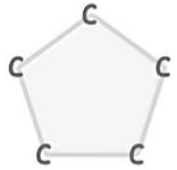


# Monosaccharide – Simple Sugar

GENERAL FORMULA: $C_n(H_2O)_n$

NUMBER FROM 3–7

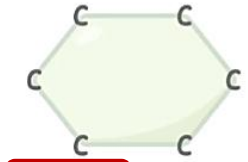
PENTOSE SUGARS



(RIBOSE & DEOXYRIBOSE)

5 CARBON ATOMS

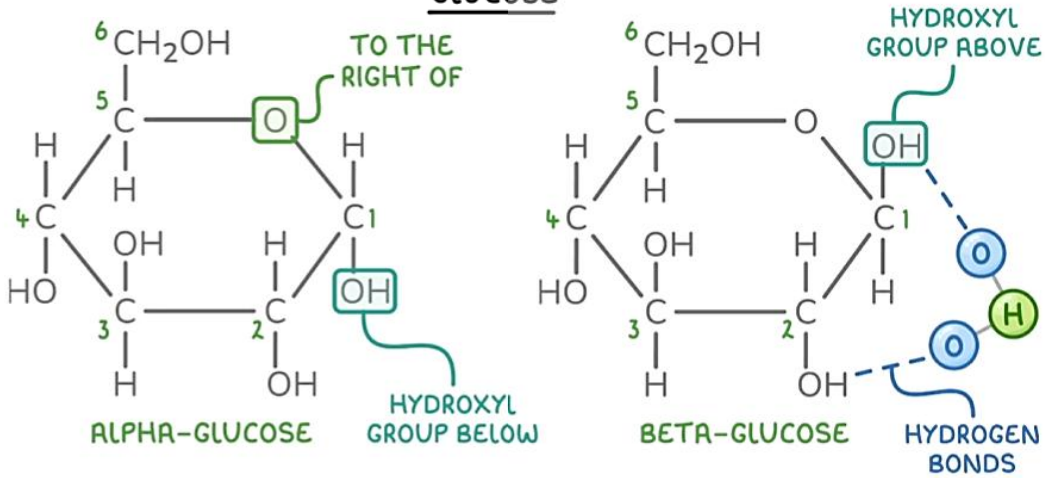
HEXOSE SUGARS



(GLUCOSE, FRUCTOSE & GALACTOSE)

6 CARBON ATOMS

GLUCOSE

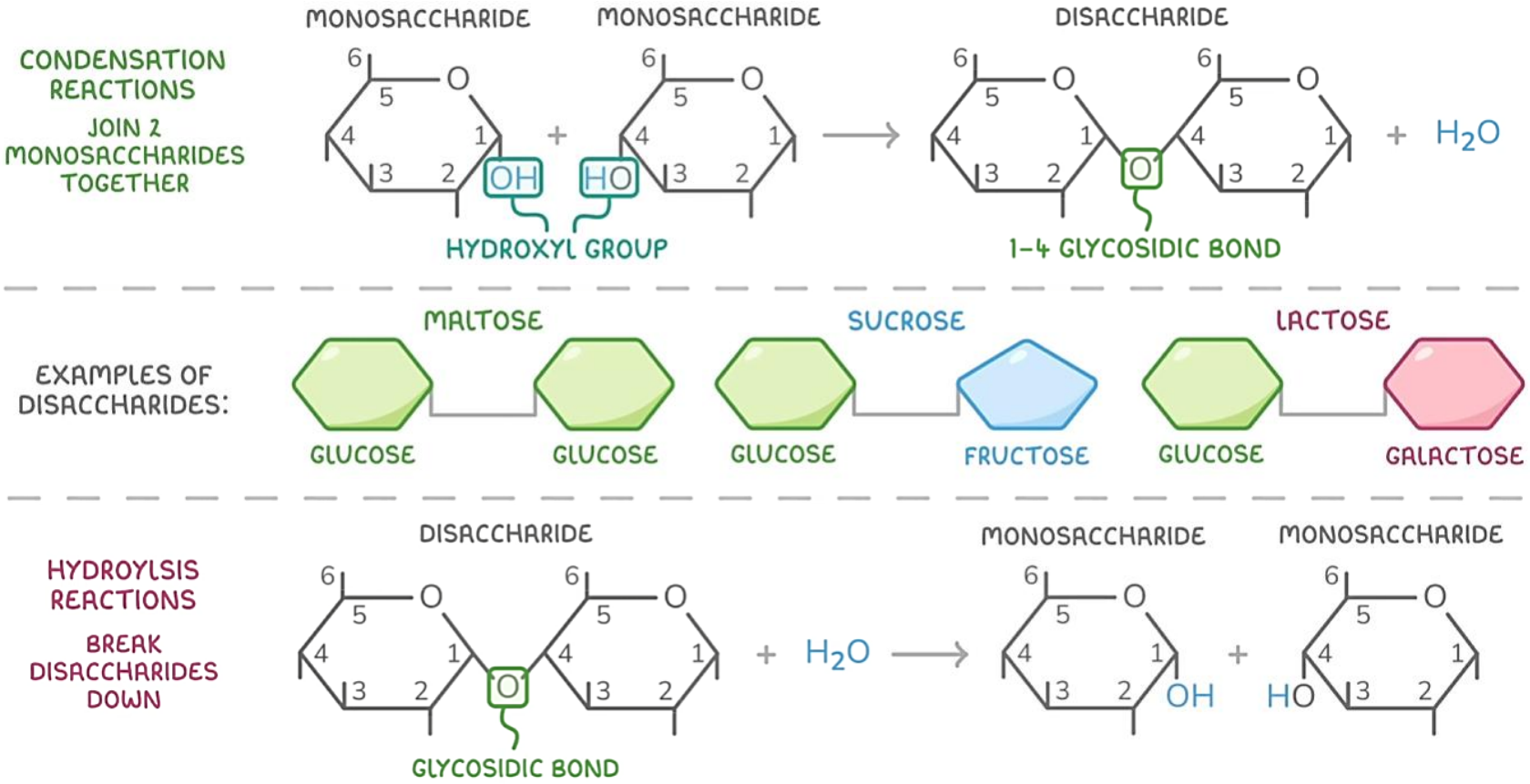



KEY FEATURES OF GLUCOSE:

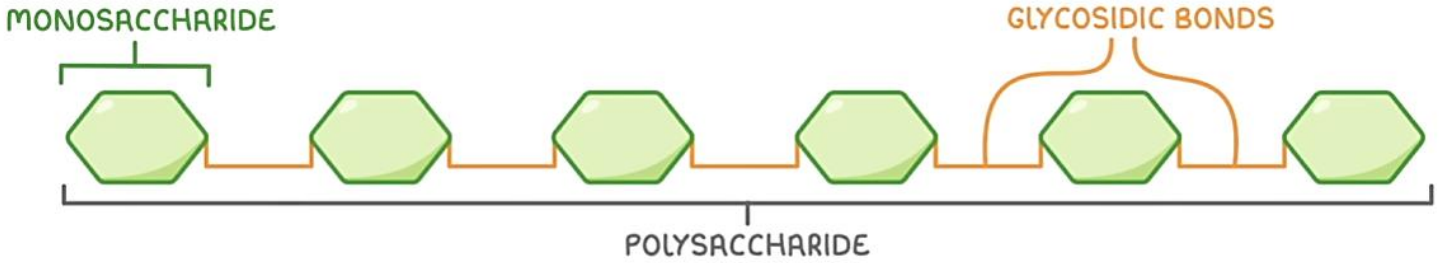
1. SOLUBLE
   - DISSOLVES EASILY IN WATER
   - IDEAL FOR TRANSPORT AROUND BODY VIA BLOOD PLASMA

2. RICH IN ENERGY
   - BONDS STORE A LOT OF ENERGY
   - RELEASED WHEN BONDS ARE BROKEN

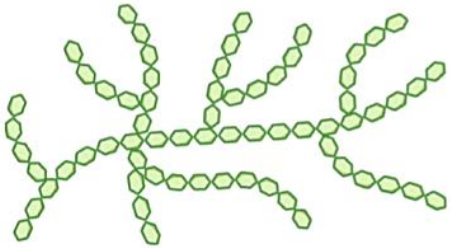
# Disaccharides

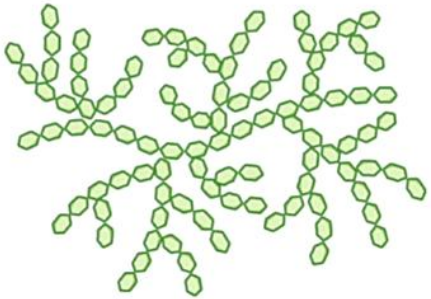


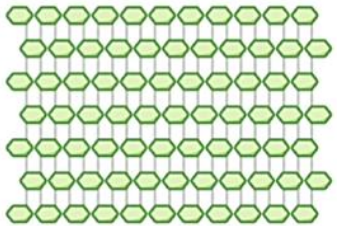
# Polysaccharides



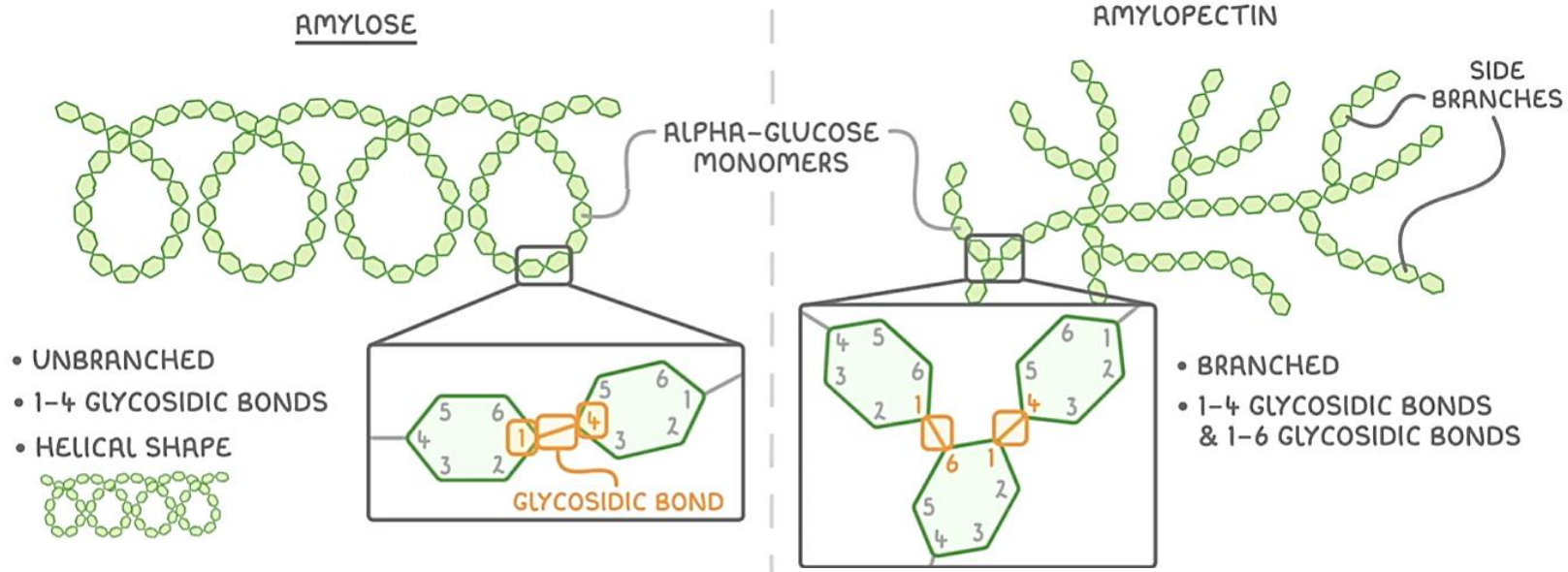

EXAMPLES OF POLYSACCHARIDES:


STARCH


GLYCOGEN


CELLULOSE

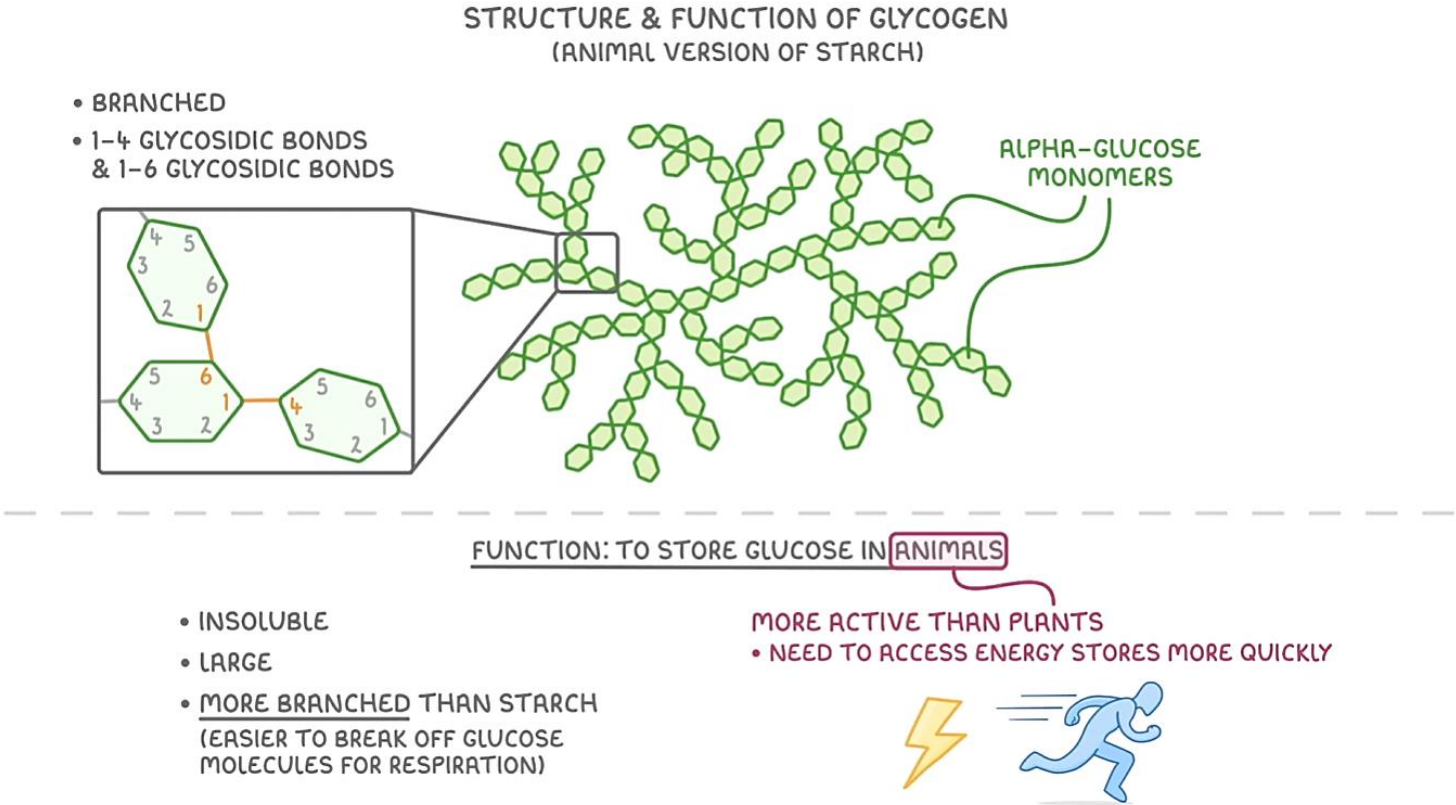
# Starch



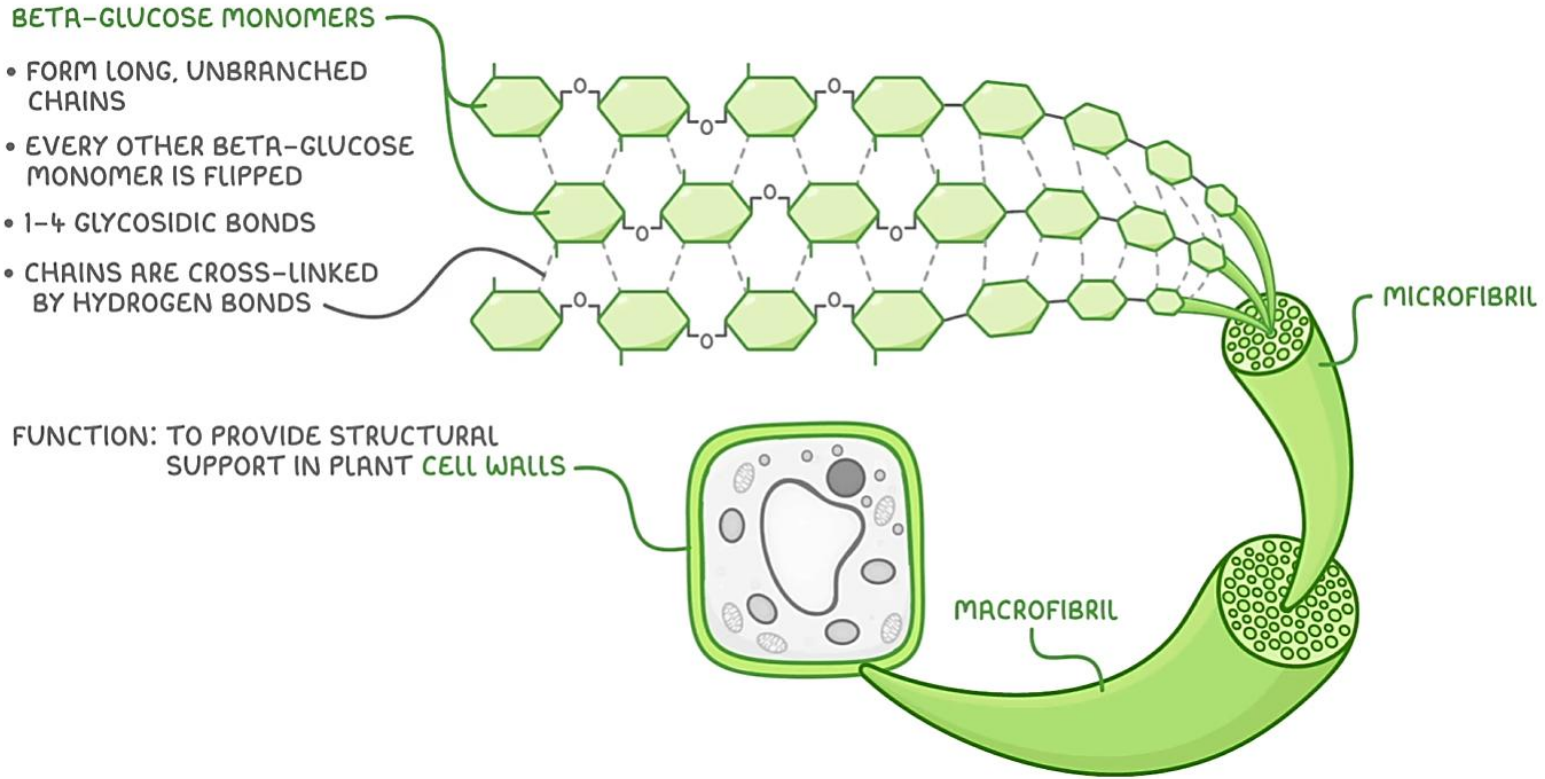

FUNCTION: TO STORE GLUCOSE IN PLANTS

- AMYLOSE IS COILED (COMPACT – PACK A LOT OF GLUCOSE INTO SMALL SPACE)
- INSOLUBLE (DOESN'T AFFECT WATER BALANCE)
- LARGE (WON'T EASILY DIFFUSE OUT OF CELLS)
- AMYLOPECTIN HAS SIDE BRANCHES (ENZYMES HYDROLYSE GLYCOSIDIC BONDS & BREAK OFF GLUCOSE FOR RESPIRATION)

# Glycogen


STRUCTURE & FUNCTION OF GLYCOGEN
(ANIMAL VERSION OF STARCH)
• BRANCHED
• 1-4 GLYCOSIDIC BONDS & 1-6 GLYCOSIDIC BONDS
ALPHA-GLUCOSE MONOMERS
FUNCTION: TO STORE GLUCOSE IN ANIMALS
• INSOLUBLE
• LARGE
• MORE BRANCHED THAN STARCH
(EASIER TO BREAK OFF GLUCOSE MOLECULES FOR RESPIRATION)
MORE ACTIVE THAN PLANTS
• NEED TO ACCESS ENERGY STORES MORE QUICKLY

# Cellulose

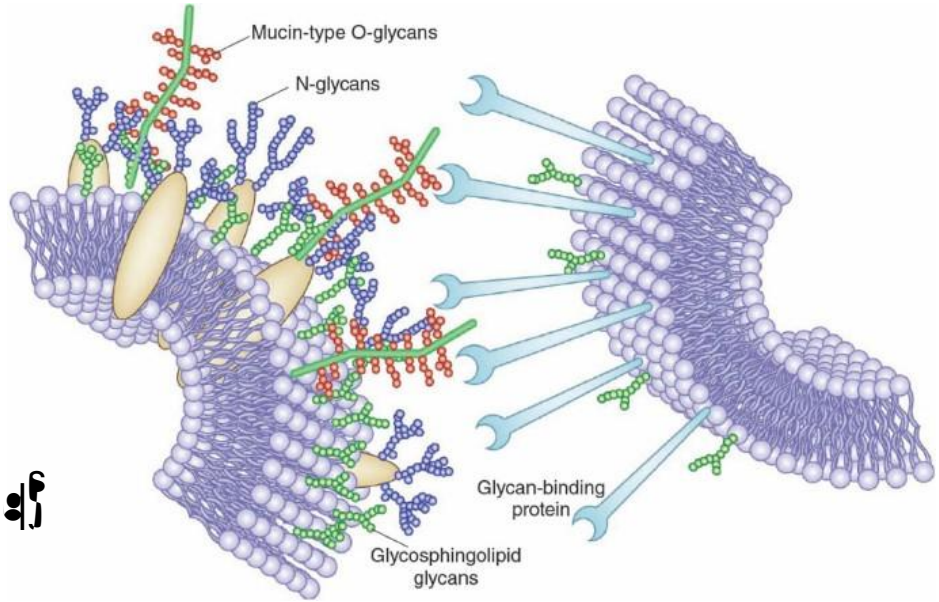

BETA-GLUCOSE MONOMERS
• FORM LONG, UNBRANCHED CHAINS
• EVERY OTHER BETA-GLUCOSE MONOMER IS FLIPPED
• 1-4 GLYCOSIDIC BONDS
• CHAINS ARE CROSS-LINKED BY HYDROGEN BONDS
MICROFIBRIL
FUNCTION: TO PROVIDE STRUCTURAL SUPPORT IN PLANT CELL WALLS
MACROFIBRIL

# Polysaccharides (the “in-between” group)

typically 3–10 monosaccharide units
technically, they can be polysaccharides
very important biologically, especially in
- cell–cell recognition
- immune signaling
- protein and lipid modification

Examples:
peptidoglycan (polysaccharide)
glycogen (polysaccharide)
**glycan chains on glycoproteins and glycolipids**

only exceptions are the polysaccharides inside the polysaccharides”, which is why they’re often grouped

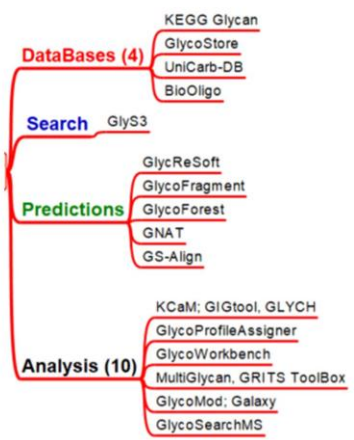


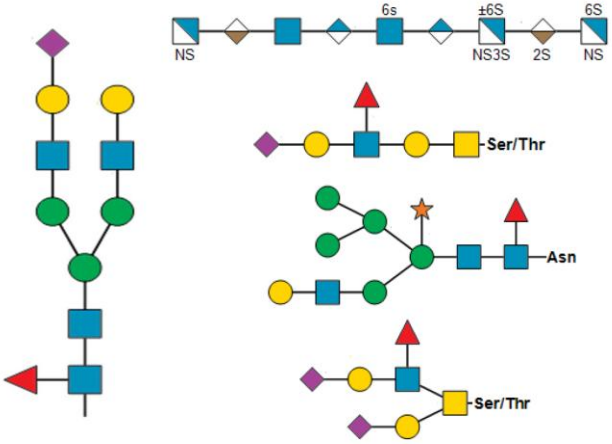
# Glycan

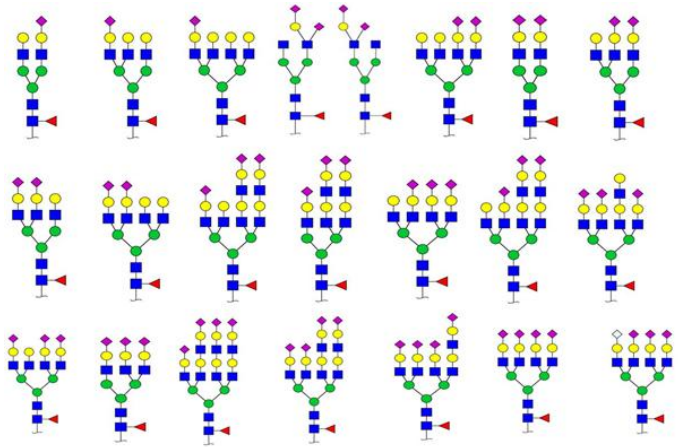
Complex sugar chains (carbohydrates) that participate in various biological processes (glycoproteins and glycolipids)

Many viruses use the recognition, binding and infection of host cells by recognizing D-glycans on the surface of host cells, and influence every step in virus infection

They also function as receptors (glycoproteins) and are the structural basis for blood group antigens (glycoproteins) and are also involved in complex functions are being uncovered



Typical native N-Linked Glycans of rhEPO expressed by CHO cell line
(Szabo et al. Journal of proteome research. 2018 Feb 16.)

# THANK YOU!